Ecosphere

Can AI predict animal movements?

Filling gaps in animal trajectories using Inverse Reinforcement Learning

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Appendix S1: Data treatment

Given the recorded positional data of a streaked shearwater, round-trip trajectories are extracted as follows. First, we determined the positions where the shearwater crossed a threshold in longitude (139.180275–139.300358) and latitude (38.411455–38.515777), around the island; those positions are the start and end of the round-trip trajectory candidates. Then, only trajectories lasting more than eight hours were accepted as round trips.

We further selected the round-trip trajectories that are adequate to train our IRL model based on the following criteria. (1) We selected trajectories that have short missing parts (less than about 50 km in length) in order to evaluate the performances of the methods by interpolating artificially created missing parts. (2) We used long-distance trajectories from the starting point (i.e., the breeding colony) to the farthest location during the round trip. A machine-learning-based algorithm requires diverse training data both in terms of quantity and quality to perform accurate prediction. In the case of trajectory interpolation, trajectories moving over various environmental locations are desirable. Therefore, we selected trajectories with distances greater than approximately 300 km from the starting location to the farthest location during the round trips. We then extracted only the outward

journey from the start to the farthest location, rather than return journey from the farthest location to the breeding colony. In RL or IRL frameworks, initial and terminal states (i.e., the start and goal, respectively) should be given. Therefore, we defined the start location and the farthest point as the initial and terminal states, respectively. Finally, we used 106 trajectories (53 males and 53 females) in the following analyses (**Figs. 2a and 2b**).